Is It the Model or the Reading?
A Fast-Weight View of ViT Attention Maps

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Abstract

Deep-layer ViT attention maps can emphasise background patches. We test how much changing the reading rule improves these maps with the model fixed. The established fast-weight view motivates three assumptions—image specificity, spatial location and output contribution—each tested with a diagnostic proxy, correction and matched control. On DeiT-S/B, existing norm weighting raises object enrichment by $0.038/0.065$ and human alignment by $0.033/0.058$; three of four gains exceed the matched control. This reading yields no improvement exceeding our practical margin for deletion faithfulness or in DINO ViT-S. Probe relocation lowers all nine deep-layer model–metric scores relative to received attention, and combining corrections improves only one of four pre-registered comparisons. These results identify an effective alignment adjustment and an unsuccessful use of index decoding for relocation; general diagnostic prediction and the cause of relocation failure remain open.

1 Introduction

Attention maps are widely used to visualise what a vision transformer (ViT) [13] attends to [1, 8, 10]. Deep-layer maps can allocate substantial attention to high-norm background tokens; registers address this by changing the model [11, 17]. We ask how much map quality can improve by changing only the *reading rule*.

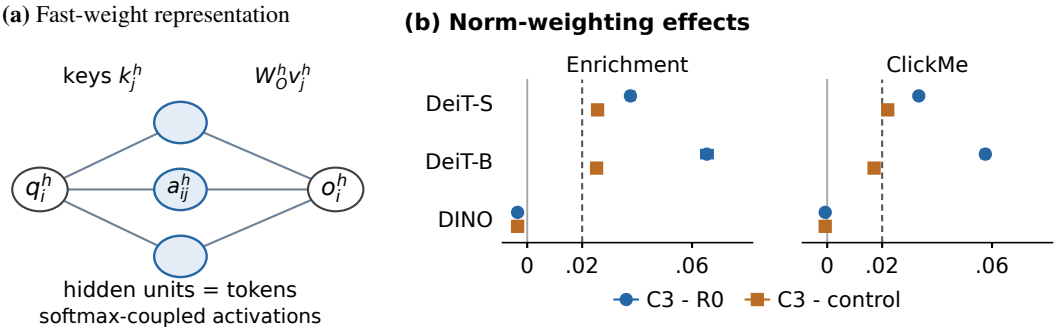


Figure 1: (a) Keys and values form image-conditioned weights; tokens are hidden units. (b) Contribution weighting (C3) minus received attention (R0) or its matched control (zero high-norm tokens), layers 9–12. Points: paired mean differences; bars: 95% intervals. The dashed line is the registered 0.02 practical margin.

Received attention sums attention arriving at each patch token. Interpreting it links activations to image-specific information, token indices to input locations, and activation magnitudes to output contributions. The established fast-weight view represents these tokens as hidden units (Fig. 1a). We organise tests of these links on frozen DeiT-S/B [25] and DINO ViT-S [8].

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(1) **A controlled diagnostic framework** linking three reading assumptions to explicit proxies, corrections and controls.
(2) **Conditions and limits of existing norm weighting:** localisation and human-alignment gains in DeiT-S/B, without a deletion-margin improvement or meaningful DINO gain.
(3) **A failed proxy-guided correction:** patch-index decoding does not yield effective score relocation, and combining corrections improves only one of four registered cells.

2 Related work

2.1 Reading attention and evaluating maps. Whether attention explains predictions is debated [16, 28]. Readings incorporate layers [1], relevance propagation [9], gradients [10], norms [18, 31] or statistical filtering [3]. Kobayashi et al. already establish that activation alone omits the transformed value magnitude. We use this existing reading to test when that omission matters. Plausible maps can ignore model parameters [2], deletion depends on occlusion [5], and CNN centre priors can confound evaluation [19]; our matched controls address specific alternative explanations.

2.2 Deep tokens and fast weights. Registers absorb high-norm background tokens [11, 17], which resemble attention heads that learn to do nothing [6]; supervision also shapes ViT attention [26]. Language-model token identity can fade [7], yet deep CLS-trained ViT tokens retain similarity to their input patches [21]. Fast weights are written by another network [24]; this perspective connects linear attention to fast-weight programming [23], softmax attention to Hopfield memory [22], and motivates efficient attention such as MiTA [27]. We use it to organise an empirical study of reading assumptions, rather than propose a new attention mechanism.

3 Method

3.1 An exact representation, an interpretive framework. In layer ℓ, normalised tokens $x_1, \ldots, x_{N+1}$ (CLS and $N = 196$ patches) enter H heads. Head h forms $q_i^h = W_Q^h x_i$, $k_j^h = W_K^h x_j$ and $v_j^h = W_V^h x_j$; its attention update is

$$o_i = \sum_h W_O^h \sum_j a_{ij}^h v_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j\big(q_i^{h\top} k_j^h / \sqrt{d}\big). \tag{1}$$

For a fixed image, each head is exactly a two-layer network in q_i^h: hidden unit j has input weights k_j^h, activation a_{ij}^h and output weights $W_O^h v_j^h$. The softmax couples hidden activations through normalisation. This algebraic representation is exact; the reading assumptions derived from it require empirical tests. Received attention is $s_j = \sum_{h,i} a_{ij}^h$, placed at patch j. Queries include CLS and all patches; maps exclude CLS and sum layers 9–12, fixed before the formal runs.

3.2 Assumptions, proxies and controls. Table 1 links three assumptions to tests: shared structure may obscure image-specific information; unit meaning requires measurement [4]; and activation omits output weights [18, 30]. These are interpretive assumptions; diagnostics remain proxies. Each correction must improve over received attention *and* exceed its matched control to support an assumption-specific benefit. Contribution weighting measures output-vector magnitudes, omitting cancellation and later computation. Probe relocation uses $P(p \mid x_j)$ trained to decode token index p, not input-content provenance.

3.3 Metrics and decision rules. A centred Gaussian correlates 0.41 with raw ClickMe maps. Two *centre-controlled* metrics address this prior: *same-ring enrichment*, the object rate of top-scoring patches minus the rate at the same radius (ImageNet-S masks [15]); and *image-specific ClickMe alignment*, Spearman correlation with the image's human map minus that with another image's [14, 20]. These controls target specific prior effects. *Deletion faithfulness* compares least- and most-important-first token-removal curves [5]. With 1,000 paired image-bootstrap resamples, a difference is *higher/lower* if its 95% interval excludes zero and its magnitude exceeds 0.02 (0.05 for deletion); otherwise it is *comparable*, a registered category rather than formal equivalence.

Table 1: Reading assumptions, diagnostic proxies and tests. Outcomes concern layers 9–12 on enrichment and ClickMe (S/B/D: DeiT-S/B, DINO). "Signal" summarises observed proxy patterns, not a validated failure classifier. †B ClickMe does not exceed its control.

Assumption	Diagnostic proxy	Correction	Matched control	Signal / benefit
Specificity	mean-map variance share	subtract mean map, $s - \bar{s}$	subtract centre Gaussian	small / none
Location	patch-index probe accuracy	relocate, $r_p = \sum_j s_j P(p \mid x_j)$	equal-displacement blur	S,B,D / none
Contribution	attention on high-norm tokens	weight, $c_j = \sum_{h,i} a_{ij}^h \lVert W_O^h v_j^h \rVert$	zero high-norm tokens	S,B / S,B†

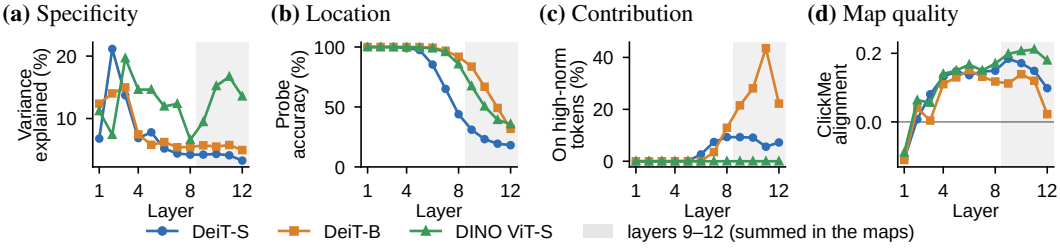


Figure 2: Per-layer diagnostic proxies and uncorrected map quality: (a) mean-map variance share; (b) patch-index probe accuracy; (c) attention on high-norm tokens; (d) image-specific ClickMe alignment. Bands: 95% intervals; shading: layers 9–12.

4 Experiments

4.1 Setup and registered predictions. We evaluate frozen supervised DeiT-S/B and DINO ViT-S/16. Model-side comparisons use a test-time register [17] and the released MiTA top-k checkpoint [27], both at size S. Data comprise 5,000 ImageNet images [12], 12,419 ImageNet-S images, 17,408 ClickMe maps and 4,000 held-out images (2,000 probe-train; 2,000 calibration/test). Metrics, margins and comparisons were fixed before formal runs (App. A). **R1:** contribution weighting in DINO and relocation at layers 1–4 remain comparable. **R2:** all three corrections improve all four supervised centre-controlled cells. **R3:** neither model-side variant is higher than DeiT-S. Single-correction/control comparisons had no registered direction; their diagnosis–benefit interpretation is post hoc.

4.2 Execution and sanity checks. Restricting attention to $k = 16$ region-selected units preserves $72/83/61\%$ of predictions in DeiT-S/B/DINO, versus $9/30/4\%$ for random units (App. B). Existing parameter-randomisation checks pass for all five readings under their registered criterion (App. F).

4.3 Diagnostic patterns (Fig. 2). The mean map explains only 3–6% of spatial variance in supervised layers 6–12, at most 17% in DINO; in layers 9–12 its structure is 82–98% radial. Index-probe accuracy declines from nearly 100% in early layers to $18/32/36\%$ at layer 12 (chance: 0.5%). High-norm tokens (block-input norm > 3 times the image median) receive up to $9/44\%$ of attention in DeiT-S/B and none in DINO. Their value norms average $0.29/0.23$ times the other patches' median in deep layers (App. B). These proxies motivate tests; they do not themselves establish misreading.

4.4 Corrections and registered outcomes (Table 2). Contribution weighting improves alignment, with limits. Enrichment rises by $0.038/0.065$ and ClickMe by $0.033/0.058$ in DeiT-S/B. Gains exceed the zero-high-norm control in three of these four cells; DeiT-B ClickMe does not. Deletion gains ($0.017/0.029$) are below the 0.05 margin. DINO differences are at most 0.004 in magnitude, satisfying the contribution part of R1. Post hoc, correctly assigned weights outperform shuffled weights in all four supervised cells; value-only weighting is comparable (App. B). Thus the useful correction is an existing norm-based reading [18, 31].

Table 2: Layers 9–12; all metrics ↑. Arrows: higher/lower versus received attention under the registered rule (model-side rows: DeiT-S). ◇: higher than the matched control. **Bold**: higher than both. Controls are indented. GAE [10] uses gradients and is a descriptive reference. Enr.: enrichment; Click.: ClickMe; Del.: deletion.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	**0.178**$^{\uparrow\diamond}$	**0.190**$^{\uparrow\diamond}$	0.253	**0.146**$^{\uparrow\diamond}$	0.165$^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	0.121$^{\uparrow}$	0.148$^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077$^{\diamond}$	0.087$^{\downarrow\diamond}$	0.144$^{\diamond}$	0.201	0.252$^{\uparrow}$	0.233$^{\downarrow}$
subtract centre Gaussian	0.133	0.168	0.167$^{\downarrow}$	0.056$^{\downarrow}$	0.062$^{\downarrow}$	0.024$^{\downarrow}$	0.200	0.253$^{\uparrow}$	0.232$^{\downarrow}$
Probe relocation	0.041$^{\downarrow}$	0.084$^{\downarrow}$	0.076$^{\downarrow}$	0.027$^{\downarrow\diamond}$	0.050$^{\downarrow}$	0.051$^{\downarrow}$	0.084$^{\downarrow}$	0.143$^{\downarrow}$	0.141$^{\downarrow}$
matched blur	0.095$^{\downarrow}$	0.152	0.181$^{\downarrow}$	−0.008$^{\downarrow}$	0.068$^{\downarrow}$	0.097$^{\downarrow}$	0.189	0.226	0.287
All three corrections	0.059$^{\downarrow}$	0.123$^{\downarrow}$	0.076$^{\downarrow}$	0.079	0.137$^{\uparrow}$	0.085$^{\downarrow}$	0.088$^{\downarrow}$	0.177$^{\downarrow}$	0.090$^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	0.100$^{\downarrow}$	0.115$^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Other corrections do not yield consistent benefits. Mean subtraction's only R0 gain (DINO ClickMe) matches its control. Relocation lowers all nine deep-layer cells and underperforms matched blur in DeiT-S and DINO. Early relocation remains comparable (R1). The combination improves only DeiT-B ClickMe: R2 *partly holds* (1/4), without consistent benefit.

Model-side results have a narrower scope. The register gains $0.012/0.013$ on enrichment and ClickMe, below the 0.02 margin. MiTA scores lower, but differs in training and preprocessing. R3 holds under the registered comparisons; these results do not establish a general advantage over model-side interventions.

4.5 Does diagnosis predict correction benefit? The post-hoc correspondence holds for contribution weighting, not relocation. Token-specific weights matter, but dose–gain associations are weak (Spearman 0.045–0.158; App. B). Local errors and the tested MLP probe narrow relocation explanations (App. C); probe/calibration limitations remain possible. Index decodability differs from content provenance: local content may persist after position codes weaken [21]. Whether that mismatch causes relocation failure remains untested.

5 Discussion

5.1 Interpretation and limitations. Reading changes can improve localisation and human alignment without improving deletion beyond the chosen margin. A declining diagnostic is therefore insufficient grounds for a correction. Evidence covers three frozen S/B classifiers and received attention only; CLS attention, larger/register-trained models and repeated training seeds are untested. The high-norm threshold was not varied. ClickMe images likely overlap model training data. Gains are modest, visually subtle (App. D) and below GAE's; diagnostic prediction beyond these checkpoints remains open.

5.2 Ethical considerations. Public images and checkpoints contain people and known biases. Improved map alignment remains partial evidence about model behaviour.

6 Conclusion

Existing norm weighting improves deep-layer localisation/alignment in DeiT-S/B; probe relocation fails and the registered combination lacks consistent gains. The fast-weight view organises these controlled tests. Validating diagnostic-guided correction requires token-to-input attribution and broader model coverage. Reading changes should be compared with matched controls, while separating alignment from prediction faithfulness.

References

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[2] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[3] Meghna P. Ayyar, Jenny Benois-Pineau, and Akka Zemmari. There is more to attention: Statistical filtering enhances explanations in vision transformers. arXiv:2510.06070, 2025.

[4] David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network dissection: Quantifying interpretability of deep visual representations. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6541–6549, 2017.

[5] Stefan Blücher, Johanna Vielhaben, and Nils Strodthoff. Decoupling pixel flipping and occlusion strategy for consistent XAI benchmarks. *Transactions on Machine Learning Research (TMLR)*, 2024.

[6] Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable transformers: Removing outliers by helping attention heads do nothing. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[7] Gino Brunner, Yang Liu, Damián Pascual, Oliver Richter, Massimiliano Ciaramita, and Roger Wattenhofer. On identifiability in transformers. In *International Conference on Learning Representations (ICLR)*, 2020.

[8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[9] Hila Chefer, Shir Gur, and Lior Wolf. Transformer interpretability beyond attention visualization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 782–791, 2021.

[10] Hila Chefer, Shir Gur, and Lior Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 397–406, 2021.

[11] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.

[12] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

[13] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[14] Thomas Fel, Ivan Felipe, Drew Linsley, and Thomas Serre. Harmonizing the object recognition strategies of deep neural networks with humans. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Shanghua Gao, Zhong-Yu Li, Ming-Hsuan Yang, Ming-Ming Cheng, Junwei Han, and Philip Torr. Large-scale unsupervised semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6), 2023.

[16] Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019.

[17] Nick Jiang, Amil Dravid, Alexei A. Efros, and Yossi Gandelsman. Vision transformers don't need trained registers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[18] Goro Kobayashi, Tatsuki Kuribayashi, Sho Yokoi, and Kentaro Inui. Attention is not only a weight: Analyzing transformers with vector norms. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.

[19] Keying Kuang, Iain Carmichael, and Elizabeth Purdom. When random saliency looks trained: Architectural center bias in CNN interpretability. In *International Conference on Machine Learning (ICML)*, 2026.

[20] Drew Linsley, Dan Shiebler, Sven Eberhardt, and Thomas Serre. Learning what and where to attend. In *International Conference on Learning Representations (ICLR)*, 2019.

[21] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[22] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.

[23] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, 2021.

[24] Jürgen Schmidhuber. Learning to control fast-weight memories: An alternative to dynamic recurrent networks. *Neural Computation*, 4(1):131–139, 1992.

[25] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, 2021.

[26] Matthew Walmer, Saksham Suri, Kamal Gupta, and Abhinav Shrivastava. Teaching matters: Investigating the role of supervision in vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7486–7496, 2023.

[27] Qishuai Wen, Zhiyuan Huang, Xianghan Meng, Wei He, and Chun-Guang Li. Mixture-of-top-k attention: Efficient attention via scalable fast weights. arXiv:2602.01219, 2026.

[28] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019.

[29] Ross Wightman. PyTorch image models. GitHub repository, `https://github.com/huggingface/pytorch-image-models`, 2019.

[30] Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, and Antonio Torralba. Learning deep features for discriminative localization. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2921–2929, 2016.

[31] Hongbo Zhu, Theodor Wulff, Rahul Singh Maharjan, Jinpei Han, and Angelo Cangelosi. Noise-free explanation for driving action prediction. In *European Conference on Artificial Intelligence (ECAI)*, volume 392 of *Frontiers in Artificial Intelligence and Applications*, 2024. doi: 10.3233/FAIA240520.

A Experimental details

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs, taken from timm [29]. *DeiT-S and DeiT-B* [25]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [17]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [27] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table 2 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side fixes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [15]: all 12,419 validation images with segmentation masks; enrichment is defined for 12,150 of them (12,144 for DINO ViT-S), after excluding images without valid foreground or background patches, and deletion uses a random subset of 2,000. *ClickMe* [14, 20]: 17,408 images with human importance maps. Images are resized from 256×256 to 224×224 without cropping; heatmaps are resized to 64×64, blurred with a 10×10 Gaussian ($\sigma = 10$), then resized to 224×224. *Held-out set*: 4,000 ImageNet validation images excluded from map-quality and execution evaluation; 2,000 train the position probes, and the other 2,000 give the mean maps, the probe accuracies and the blur widths. While building this set we found that only 2 of 4,002 candidate validation images matched a ClickMe image, so the ClickMe images are very likely ImageNet training images, which all three models have seen.

Diagnostics. All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and Var_j the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images x. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Contribution*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

Corrections and matched controls. All are applied per layer and then summed over layers 9–12; queries run over all 197 tokens and maps cover the 196 patches. *Mean-map subtraction* subtracts the model's mean received-attention map over the 2,000 held-out images. Its control subtracts a centred isotropic Gaussian ($\sigma = 3.5$ patches) scaled to the spatial standard deviation of that mean map; it is exactly radial. *Probe relocation* uses one linear 196-way classifier per model and layer (36 in all) on the standardised tokens after the block's first layer norm (Adam, learning rate 0.01, 30 epochs) and computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, which keeps the total score. Its control is a Gaussian blur whose width is set so that it moves the score as far, in mean squared displacement, as relocation does on held-out tokens of that layer (0.7–2.9 patches). *Contribution weighting* computes $c_j = \sum_h \sum_i a_{ij}^h \|W_O^h v_j^h\|_2$, where W_O^h is the head's slice of the output projection; $a_{ij}^h \|W_O^h v_j^h\|$ is the per-head norm of Kobayashi et al. [18]. A *high-norm token* is a patch whose block input has a norm above three times the image's median patch norm in that layer; the control sets received attention to zero at these patches. *All three corrections*: contribution weighting, then probe relocation, then subtraction of the held-out mean of the resulting map.

Metrics. *Same-ring object enrichment.* A patch is valid if it contains at least one non-ignored pixel, and is an object patch if at least half of its non-ignored pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among

all its patches, and average over rings, weighted by their selected patches. Invalid patches are excluded from selection and ring baselines. Ties are broken with the fixed seeded perturbation used in the original runs. A purely radial map scores zero in expectation over tie-breaking. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the previous image's human map within each batch (`roll(1)`; batch size 64). For an image-independent map, these differences average to zero over the paired set because the control permutes the same human maps; individual-image differences need not be zero. This removes fixed-prior alignment on average, without guaranteeing invariance to a prior mixed into an image-dependent map. *Deletion faithfulness.* Patches are removed in the order of the map by dropping their tokens; at budgets of 0 to 147 patches we record whether the top-1 class is unchanged. The score is the symmetric relevance gain of Blücher et al. [5]: the area under the least-important-first curve minus that under the most-important-first curve. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps have exactly zero mean on the latter, up to numerical precision); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired bootstrap over images (1,000 resamples, 95% percentile intervals). A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its size exceeds 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. The registered rule and margins are unchanged: comparable combines intervals containing zero with effects below the margin and is not a formal equivalence claim. Intervals are not adjusted for multiple comparisons; the practical margin does not itself provide multiplicity control.

Registered predictions. Before the formal runs we fixed the definitions above, the margins, the comparisons of Table 2 and the predictions of Table A.1; the record is the file `docs/preregistration.md` of the repository. R2 was registered with three possible verdicts: holds (four of four cells), partly holds (one to three) and does not hold (none). For each correction against its control, the comparison was fixed without a direction. Earlier runs of the execution test and of the diagnostics had been seen, so the last two rows are replications. Other analyses are marked post hoc.

Table A.1: Registered predictions and outcomes (centre-controlled metrics: enrichment and ClickMe alignment).

	Prediction	Outcome
R1	Contribution weighting is comparable to received attention in DINO ViT-S on all three metrics	holds (3 of 3 cells)
R1	Relocation at layers 1–4 is comparable to received attention on the centre-controlled metrics in all models	holds (6 of 6 cells)
R2	All three corrections together are higher than received attention on the centre-controlled metrics in DeiT-S and DeiT-B	partly holds (1 of 4 cells: DeiT-B, ClickMe)
R3	Neither model-side change is higher than DeiT-S received attention on the centre-controlled metrics	holds (4 of 4 cells)
–	Execution agreement rises with k and exceeds that of random sets at every k	holds in all models
–	Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S	holds in all models

Compute and code. All experiments took about 19 GPU hours on NVIDIA L4 GPUs. Code and result tables: `https://github.com/hhuang999/vit-fastweight-lens`.

B Additional results

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer and head, each region's mean query is scored against all keys, and its k highest-scoring keys form the region's reading set (the CLS query keeps its own top-k set). The sets come from one pass of the unmodified model. We then rerun the model with every query restricted to its region's set, in all 12 layers at once, and record whether the top-1 class is unchanged. Figure B.1 shows that top-k sets beat random sets of the same size at every k from 4 to 128 in all three models, and that agreement rises with k. The dotted line alone was run with a crop ratio of 0.875 for all models (0.9 for DeiT-S and DeiT-B elsewhere).

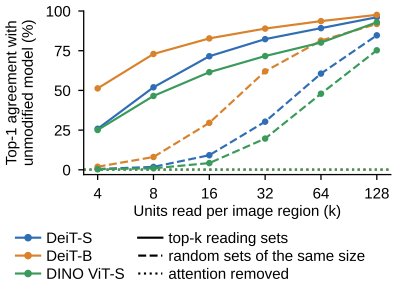


Figure B.1: Top-1 agreement with the unmodified model when queries read only top-k units (solid), random units (dashed) or nothing (dotted). 95% intervals: at most ±1.5 points.

Diagnostics per layer. Table B.1 lists the values plotted in Fig. 2. Two numbers of the main text are not in the table. First, in layers 9–12, 82–98% of the variance of the mean map is explained by the distance from the grid centre. Second, averaged over layers 9–12, the mean value-vector norm $\|v_j\|$ of high-norm tokens is 0.29 (DeiT-S) and 0.23 (DeiT-B) times the median of the other patches (measured on 11,919 ImageNet-S images using uniform crop ratio 0.875, unlike the main DeiT evaluation at 0.9).

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (Fig. 2d). Applied to layer 12, contribution weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Table B.1: Diagnostics per layer, as plotted in Fig. 2 (S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%).

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Differences behind Table 2. Table B.2 gives the paired differences with their intervals. The values in Table 2 have 95% intervals of at most ±0.005 on enrichment and ClickMe alignment and ±0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [-0.005, 0.004] on deletion for the test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training.

Tests of the contribution account. Contribution weighting should work by down-weighting tokens that receive attention but contribute little. One consequence was registered (R1), and two

Table B.2: Paired differences, layers 9–12: mean with 95% paired-bootstrap interval and verdict (*h* higher, *c* comparable, *l* lower). Top: against received attention. Bottom: each correction against its matched control.

Model	Same-ring enrichment	ClickMe alignment	Deletion faithfulness
Contribution weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Contribution weighting − zero high-norm tokens (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

further tests were added post hoc. *No practically meaningful gain in DINO (R1).* In DINO ViT-S, where no attention falls on high-norm tokens, every difference from received attention is at most 0.004 in size (Table B.2). *More high-norm attention, more gain (post hoc).* Across images, gains have a weak positive association with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *The weights must match the tokens (post hoc).* Against the same weights shuffled among the patches of each head, contribution weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. Finally, also post hoc, weighting by $\|v_j^h\|$ instead of $\|W_O^h v_j^h\|$ is comparable in every cell (differences -0.010 to 0.000): in practice the correction is the value-norm weighting of Kobayashi et al. [18] and Zhu et al. [31].

C Why does probe relocation fail?

Relocation lowers every deep-layer cell (Table 2). Two planned comparisons narrow the problem. Where the probe is accurate, relocation is harmless (R1): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. And in DeiT-S and DINO ViT-S it is lower than a blur that moves the score equally far (Table B.2). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none meets its specified criterion. Most probability lies near the true patch, and replacing the linear probe with the tested MLP does not restore map quality. This does not exclude all probe-capacity or calibration limitations. The probe targets index decodability rather than input-content provenance; whether this mismatch explains the failed relocation remains untested (Section 4).

Table C.1: Four explanations for the failure of probe relocation, and their tests (post hoc). Cells are the two centre-controlled metrics in the three models.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

D Example maps

The stored gallery holds 24 ClickMe images drawn at random with a fixed seed. Fig. D.1 shows the two with the smallest ClickMe index (the original v8 main-text examples), and Fig. D.2 the next four. The image-selection rule is unchanged. Received attention and contribution weighting look much alike, while probe relocation spreads the score over many more patches. The gains in Table 2 are averages of small changes in how each map ranks its patches, which single maps do not show.

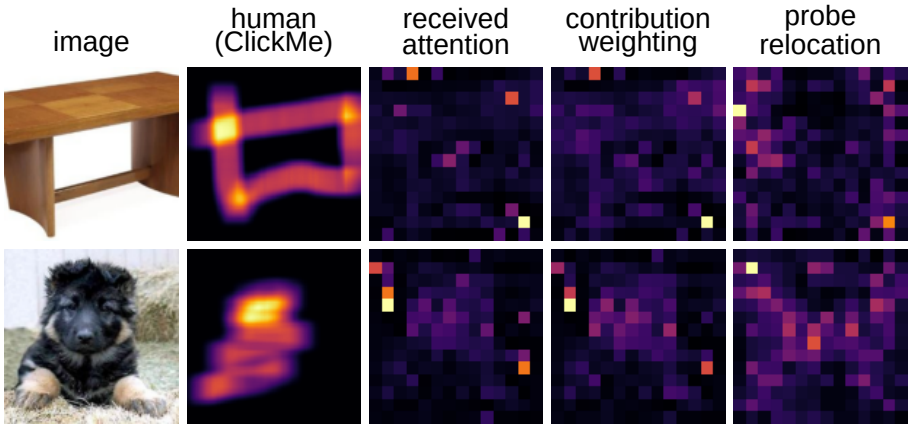


Figure D.1: The first two fixed-seed examples, DeiT-S, layers 9–12. Each map is min–max normalised; these images were not selected for correction gain.

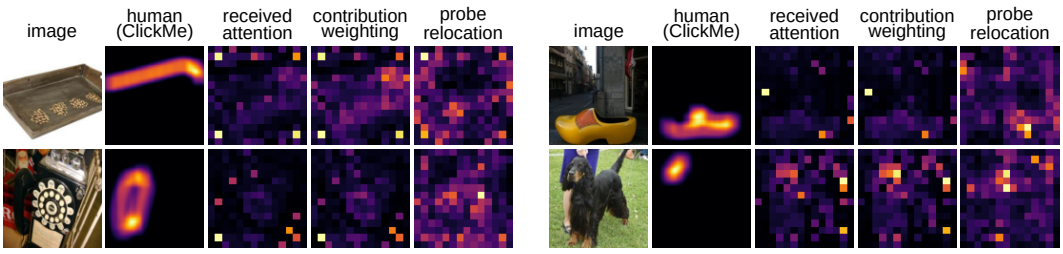


Figure D.2: DeiT-S on four further ClickMe images, layers 9–12; each map is min–max normalised.

E Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,N,N] attention a_ij^h (queries i, keys j); v: [B,H,N,d] values; w_o: [D,H,d] W_O by head
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)                # W_O^h^T W_O^h
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v).clamp_min(0).sqrt()
    received = p.sum(2)                                          # sum over queries i
    s = received.sum(1)[:, prefix:]                              # received attention s_j
    c = (received * length).sum(1)[:, prefix:]                   # contribution weighting c_j
    return s, c

def relocate(s, probs):                  # probe relocation r_p = sum_j s_j P(p | x_j)
    return torch.einsum('bj,bjp->bp', s, probs)
```

F Historical parameter-randomisation check

The original study included a model-parameter randomisation check [2] on 1,000 fixed-seed ImageNet-S images. Parameters were cumulatively re-initialised from the classifier and final norm through blocks 12–1 to the patch/position embeddings and CLS token. The trained model's probes and mean maps remained fixed. Received attention and its corrections are class-agnostic and therefore do not depend on classifier parameters alone. Table F.1 reports the stored median per-image Spearman correlations between trained and fully randomised maps, with layers 9–12 summed. All five readings meet the registered median < 0.5 criterion; the model-independent centre Gaussian fails as designed. This is a necessary sensitivity check, not proof of prediction faithfulness. The existing summary is reproduced here without rerunning the experiment.

Table F.1: Stored model-randomisation results: median map correlation after full randomisation. The five readings pass the registered < 0.5 criterion in every model.

Reading	DeiT-S	DeiT-B	DINO ViT-S
Received attention	−0.033	0.055	0.071
Mean-map subtraction	0.187	0.366	0.100
Probe relocation	0.063	0.004	−0.002
Contribution weighting	−0.064	0.070	0.114
All three corrections	0.069	0.078	0.025
Centre Gaussian	1.000	1.000	1.000